Loan Approval Analysis

Comprehensive analysis of 1,000+ loan applications exploring approval patterns, risk factors, and revenue opportunities across customer segments.

Made with GAMMA

Dataset Overview

Applicant Profile

- Age & employment status
- Annual income
- Credit score

Loan Details

- Loan amount & purpose
- Interest rate
- Approval status (1/0)

Financial Health

- Existing debts
- Debt-to-income ratio
- Credit history length

Made with GAMMA

Python Data Preparation

Workflow Steps

01

Data Loading

Imported dataset using pandas for cleaning and analysis

02

Initial Exploration

Checked data types, null values, and column structure with df.info()

03

Feature Engineering

Created credit_tier, income_bucket, and age_group columns

04

Column Reorganization

Re-arranged columns for better relatability and analysis

Credit tiers: Low (<580), Fair (580-670), Good (670-740), Excellent (740+)

Made with GAMMA

Highest Approval Rates by Segment

Customer segments with the strongest loan approval performance:

74.8%

Older Adults

Employed (8,023 customers, 6,004 approved)

74.3%

Older Adults

Self-Employed (2,317 customers, 1,722 approved)

30.7%

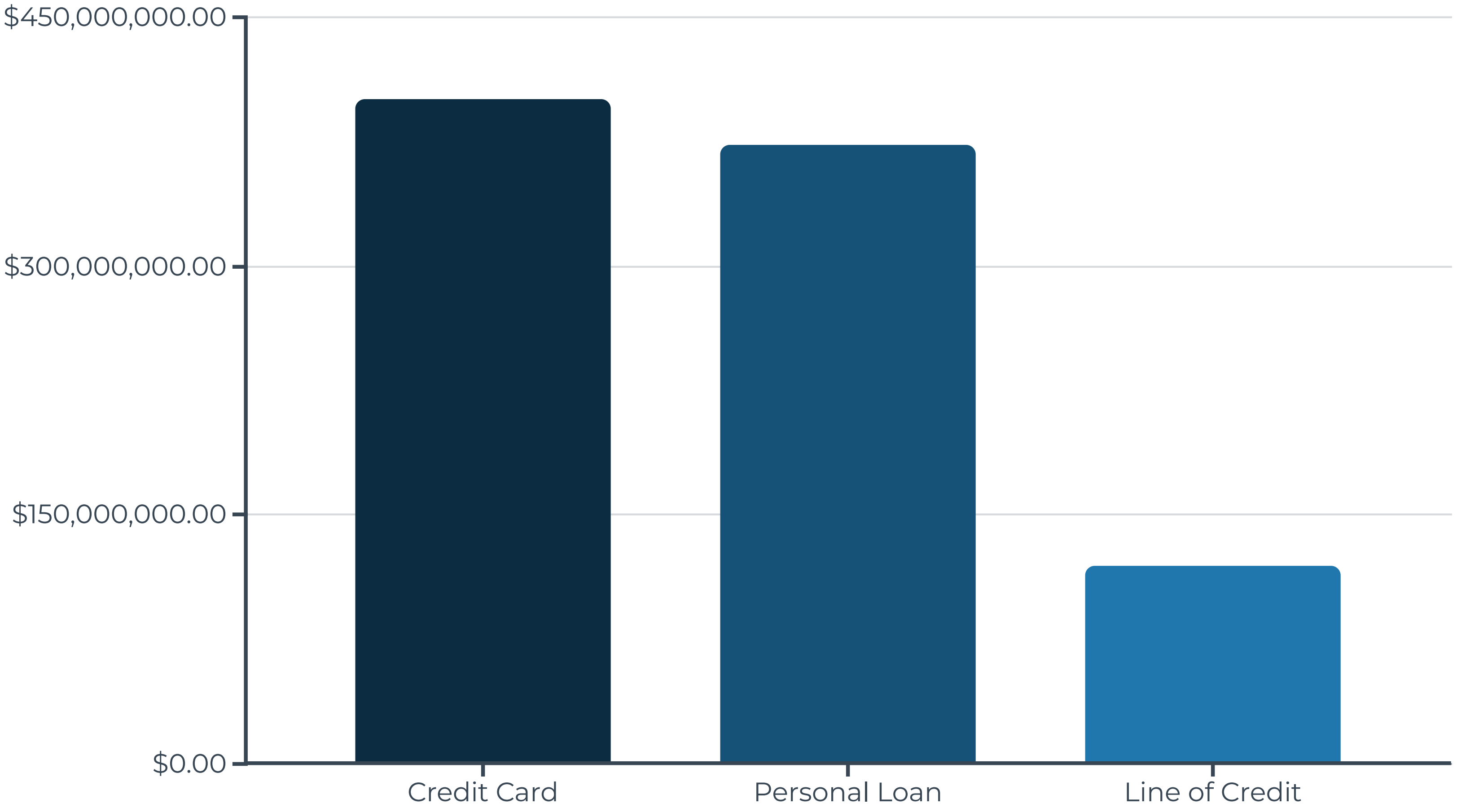
Young

Employed (8,920 customers, 2,734 approved)

Older adults show significantly higher approval rates across all employment types, while younger employed customers face the lowest approval rates.

Made with GAMMA

Revenue by Product Type



Credit Cards generate the highest approved loan volume at **$400M**, followed by Personal Loans at $373M. Average loan amounts: Credit Card ($29K), Personal Loan ($44K), Line of Credit ($22K).

Made with GAMMA

Credit Score Performance

88.4%

Excellent

3,116 of 3,526 approved

79.6%

Good

10,967 of 13,779 approved

50.2%

Fair

12,368 of 24,657 approved

13.3%

Low

1,072 of 8,038 approved

Excellent credit scores achieve the highest approval rates, while low credit tiers face significant rejection.

Made with GAMMA

Default Risk Analysis

Lowest Risk: Home Improvement

5.1% default rate (382 of 7,453 customers)

Business Loans

5.2% default rate (387 of 7,469 customers)

Highest Risk: Education

5.5% default rate (561 of 10,134 customers)

Income matters: High-income customers default at 4.4%, while low-income customers default at 6.1%. Young customers show the highest default rate at 7.4%.

Made with GAMMA

Interest Rate Patterns

Average Interest Rates by Segment

16.6%

Young Students

16.5%

Young Employed

15.8%

Young Adults Employed

14.4%

Older Adults

Pricing Leakage Alert: Some high-risk customers (low credit scores, multiple delinquencies) are receiving interest rates as low as 13%, indicating potential underpricing.

Made with GAMMA

Key Opportunities

Underpriced High-Risk Customers

Customers with low credit scores (546-562), multiple delinquencies (3-6), and defaults receiving rates as low as 13%. Opportunity to adjust pricing models.

Untapped Borrowing Capacity

Approved customers with low debt-to-income ratios represent opportunities for additional lending and cross-selling.

Revenue Optimization

Employed customers generate highest revenue with 0.25 payment-to-income ratio. Focus marketing on this segment for maximum returns.

Made with GAMMA